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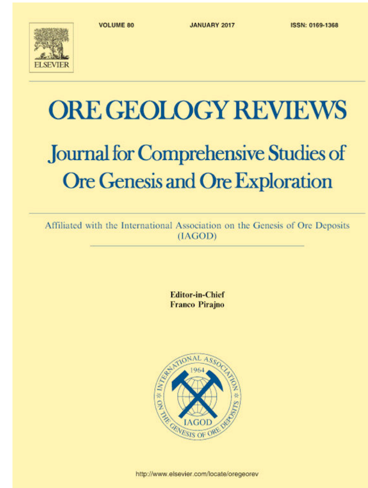

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LA-ICP-MS analysis of sulfides from the Sinongduo Ag-Pb-Zn deposit, Tibet: Insights into element incorporation mechanisms and ore genesis

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Abstract: As one of the most significant Ag-Pb-Zn deposits in the northern Gangdese polymetallic belt, the Sinongduo has been widely concerned since its discovery and hosts proven reserves of 148,136 t Pb (1.95 %), 205,722 t Zn (2.70 %), and 400 t Ag (52.33 g/t). Despite its economic importance, the mechanisms of Ag incorporation in sulfides and deposit genetic type remain debated. This study mainly employs in *sit*u LA-ICP-MS trace element analysis of sulfides (sphalerite, pyrite, and galena) to address the above issues. Three mineralization stages are identified: (I) pre-ore pyrite + quartz stage, (II) sphalerite + galena + pyrite + minor chalcopyrite + Ag-bearing minerals main stage, and (III) post-ore calcite + quartz stage. LA-ICP-MS depth profiles and multivariate statistical analysis reveal trace elements are primarily hosted as lattice-bound solid solutions and nanoscale inclusions, with limited micro-inclusion formation. In the Sinongduo deposit, where silver is the most critical element, its substitution mechanisms exhibit significant differences among sphalerite, pyrite, and galena. In sphalerite, silver incorporates into the crystal lattice via coupled substitutions involving the replacement of Zn by Fe, Cu, Sb, Ga, and Sn. In pyrite, silver enters the lattice through coupled substitutions where Fe is replaced by As, Sb, and Co. In galena, silver is incorporated into the lattice through coupled substitutions where Pb is replaced by Sb and As. Trace element thermometry (GGIMFis geothermometer) yields ore-stage temperatures of 226.0–288.0 °C (mean = 256.8 °C), consistent with fluid inclusion data. The trace element distributions in the Sinongduo sulfides, such as Fe, Mn, Cu, Ag, Sn, and Pb enrichment and Ga, Ge, In, and Co depletion in sphalerite, together with the supports from the host rock, alteration, texture and mineral assemblages, suggest that the Sinongduo deposit is a typical epithermal deposit. This study establishes the Sinongduo deposit as a typical epithermal system and demonstrates that in *situ* LA-ICP-MS analysis of sulfide trace elements provides robust constraints on deposit genesis.

Keywords: Trace elements; In *situ* LA-ICP-MS; Sulfides; Silver incorporation

mechanism; Sinongduo

1. Introduction

The Gangdese Metallogenic Belt (GMB) in the Tibetan Plateau is a globally significant polymetallic domain (Tang et al., 2009, 2021, 2024; Lang et al., 2020; Li et al., 2022; Hou et al., 2024; Wang et al., 2024), hosting Miocene porphyry Cu-Mo systems and Paleocene-Eocene Fe-Cu-Pb-Zn-Ag-Mo-W skarns (Zheng et al., 2015; Xu et al., 2024). Within the GMB, the northern Gangdese polymetallic belt (NGPB) is a key Pb-Zn-Ag-Fe-Cu-Mo producing area (Zhang et al., 2023; Ran et al., 2023), where over 20 Pb-Zn-Ag deposits are found, many of which are skarn deposits (Zheng et al., 2015). Notable Pb-Zn-Ag deposits in the NGPB include large and medium-sized ones such as Yaguila (Xu et al., 2019; Li et al., 2023), Dongzhongla (Sun et al., 2020), Mengya'a, Longmala (Jiang et al., 2023), Lietinggang, Leqingla (Zhang et al., 2019; Ma et al., 2022), Xingaguo (Tang et al., 2020), Narusongduo (Zhou et al., 2020), Chagele (Gao et al., 2021), and Longgen (Gao et al., 2020) Pb-Zn deposits.

The NGPB is one of the most important sources of lead, zinc, and silver resources in China. The Sinongduo deposit is the first epithermal Ag-Pb-Zn deposit discovered in the NGPB, with metal reserves of 148,136 tons (t) of Pb, 205,722 t of Zn, and over 400 t of silver, and average grades of 1.95% Pb, 2.70% Zn, and 52.33 g/t Ag, respectively (Tang et al., 2016). Many studies have been conducted on various aspects of this deposit, including the geological features (Qian et al., 2013; Tang et al., 2016), the petrogenesis of ore-related volcanics and granites (Ding et al., 2017; Yang et al., 2021), fluid sources and their evolution (Li et al., 2019), ore mineralogy (Meng et al., 2016; Li et al., 2017), stable isotope and ore-forming material sources (Fu et al., 2017), alteration and zonation (Huang et al., 2017; Guo et al., 2018, 2019, 2020), tectonic setting (Wang et al., 2010), and geochronology (Zhang et al., 2018; Yang et al., 2020, 2021). Previous detailed microscopic identification and electron probe analysis found that there were a large number of independent silver minerals (e.g., argentite, pearceite, pyrargyrite) in the Sinongduo deposit (Li et al., 2017). Ag can form both independent silver minerals, but can also enter the lattice of sphalerite, pyrite, and galena (Cabri et al., 1985; Cook et al., 2011; Wu et al., 2020). However, the occurrence and substitution mechanisms of Ag within the sulfides in the Sinongduo deposit is not clear, which seriously affects the efficient and comprehensive utilization of resources and the understanding of mineralization. At present, the laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) in-situ microanalysis has become a widely adopted technique for precisely determining the elemental compositions and substitutional forms in sulfides (Ye et al., 2011; Chu et al., 2022; Zhou et al., 2022; Yang et al., 2022; Xiang et al., 2024). Considering the compositional variability of sulfides influenced by factors such as temperature and pressure (George et al., 2015; Keith et al., 2020; Wu et al., 2023), the Sinongduo deposit presents an ideal hydrothermal system for investigating Ag behavior in sulfides. Additionally, various studies have proposed different genetic types for the Sinongduo deposit, such as the magmatic-hydrothermal vein type (Li et al., 2010), the stacked and reshaped type (Wang et al., 2012), the hydrothermal vein-skarn type (Zhang et al., 2018), and intermediate-low sulfidation epithermal type (Tang et al., 2016; Yang et al., 2020). The genetic classification of the Sinongduo deposit remains contentious, necessitating further research for a comprehensive understanding of its genesis. The trace elements in sulfides can offer significant insights into the genetic origins of mineral deposits and are commonly utilized as a basis for distinguishing different genetic types of these deposits (Frenzel et al., 2016; Fan et al., 2020; Xing et al., 2022).

This study mainly provides a comprehensive trace element analysis of the ore sulfides, specifically sphalerite, pyrite, and galena, found in the Sinongduo deposit. We aim to identify the distribution features and substitution mechanisms of trace elements, determine the mineralization temperature, and constrain the genetic type. This study will enhance insights into the origins of the Sinongduo Ag-Pb-Zn mineralization and offer crucial guidance for Ag-Pb-Zn exploration efforts within the GMB.

2. Regional Geology

The Qinghai-Tibet Plateau formation is mainly due to the Cenozoic India–Asia collision (Fig. 1a; Mo et al., 2005; Replumaz et al., 2010; Zhu et al., 2015; Hou et al., 2024). It consists of E-W trending terranes such as the Himalaya, Lhasa, Qiangtang, Songpan-Ganzi, and Kunlun from south to north (Fig. 1b; Yin and Harrison, 2000). The Lhasa terrane is a large block with a continental crust thickness of 65–75 km (Molnar et al., 1998), bounded by the Yarlung-Zangbo Suture Zone (YZSZ) to the south and the Bangong-Nujiang Suture Zone (BNSZ) to the north. It can be further divided into southern, central, and northern subterranes separated by the Shiquan River-Nam Tso Mélange Zone (SNMZ) and the Luobadui-Milashan Fault (LMF) (Zhu et al., 2013). The southern and northern Lhasa subterranes mainly consist of juvenile basement rocks and Mesozoic-Cenozoic sedimentary and volcanic rocks, such as the Yeba, Zenong, and Duoni Formations (Zhu et al., 2011; Pan et al., 2012). The central region preserves not only Mesozoic-Cenozoic geological records but also an ancient basement (Zhu et al., 2013). Since the Early Jurassic, the Lhasa terrane has undergone significant tectonic and magmatic activity. These activities include the Jurassic subduction of the Neo-Tethyan oceanic slab, forming the Xigaze forearc basin and the Gangdese magmatic arc batholith (Wang et al., 2008; Ji et al., 2009), and the prolonged Indian-Asian continental collision, which led to the emplacement of large arc batholiths (e.g., Paleocene syn-collisional granitoids, Eocene granitoid intrusions, basaltic subvolcanic rocks) and extensive eruptions of the Linzizong volcanic successions (LVS) (Mo et al., 2007; Zhu et al., 2015; Zhang et al., 2023). The LVS stretches over 1200 km east-west and covers >50% of the GMB's outcrop area. According to geochronological data, the LVS is divided into the Lower Dianzhong Formation (69-61 Ma), middle Nianbo Formation (61-54 Ma), and upper Pana Formation (54-44 Ma) (Mo et al., 2008; Lee et al., 2012; Chen et al., 2014; Zhu et al., 2015).

The Lhasa terrane hosts two major mineralization belts: the Gangdese porphyry copper belt (GPCB) in the southern Lhasa subterrane and the NGPB in the central Lhasa subterrane (Hou et al., 2015; Zheng et al., 2015; Zhang et al., 2019). The NGPB, extending over 400 km from Yaguila in the east to Lazong in the west along the Luobadui-Milashan Fault (LMF), contains various deposits, including Paleocene syn-collisional skarn Pb-Zn-Ag-Fe, porphyry Cu-Mo-W, hydrothermal vein Pb-Zn-Ag, and cryptoexplosive breccia Pb-Zn-Ag deposits. These deposits are genetically linked to N-dipping thrust faults (Zheng et al., 2015; Zhang et al., 2022; Zhang et al., 2023). The southeastern NGPB primarily hosts porphyry/skarn Cu-Mo-W deposits, such as the Bangpu porphyry Mo-Cu, Sharang porphyry Mo, and Hahaigang skarn Mo-W deposits. Skarn Fe-Cu deposits, like Jiaduopule, Jialapu, and Chengbapu, are mainly found in the southern margin of the central subterrane and are associated with syn-collisional granodiorites. The economically significant skarn Ag-Pb-Zn and Pb-Zn-Fe-Cu deposits in the NGPB are related to granites (Zheng et al., 2015). The ore belt can be further divided into two sub-belts: the eastern Nyenchen Tanglha Mountains Cu-Pb-Zn-Ag sub-belt, which includes deposits such as Yaguila, Sharang, Dongzhongsongduo, Hahaigang, Mengya'a, and Tangbula, and the western Ag-Pb-Zn sub-belt, which contains deposits like Narusongduo, Sinongduo, Qiagong, and Lazong (Fig. 1c) (Wang et al., 2017; Zhou et al., 2020; Yang et al., 2020).

The Sinongduo deposit is located in the southern part of the NGPB within the central Lhasa subterrane. In this area, the exposed strata mainly consist of the Lower Carboniferous Yongzhu Formation, Upper Carboniferous Laga and Angjie Formations, Lower Permian Xiala Formation, Paleocene Dianzhong and Nianbo Formations, and Quaternary. Among these, the Dianzhong Formation is the most widely distributed ore-bearing stratum at Sinongduo (Ran et al., 2023). Moreover, the magmatic rocks intruded in this region mainly contain the Paleocene rhyolite porphyry, dacite, tuff, and granite porphyry, with minor Miocene biotite granite porphyry. The radial faults and reverse thrust structures are the leading regional structures (Ding et al., 2017).

3. Ore deposit geology

The Sinongduo deposit, located in Xietongmen County's southern part of the central Lhasa subterrane (29°58′07″-29°58′57″N, 88°34′10″-88°34′58″E) (Fig. 1c), is a typical hydrothermal Ag-Pb-Zn deposit in the NGPB. The exposed stratigraphic sequences in the Sinongduo deposit mainly consist of the Paleocene Dianzhong Formation of LVS, with U-Pb ages of 65–62 Ma (Ding et al., 2017), and Quaternary gravel (Fig. 2a). The Dianzhong Formation volcanic, the main ore-hosting unit, are composed of rhyolite porphyry, volcanic breccia, crystal tuff, and dacite. The lithofacial zonation (rhyolite porphyry-volcanic breccia–crystal tuff (tuff)–dacite) indicates that a potential volcanic system exists in the Sinongduo deposit (Yang et al., 2020). The Quaternary gravel is composed primarily of unsorted rock fragments, with a maximum thickness of 35 meters. The west-east trending radial faults, southwest-northeast trending fault, and southwest-northeast trending cracked zones are well developed following the volcanic circle (Fig. 2). The cracked zones represent the principal hosts for the vein-type ores in the Sinongduo deposit. The Paleocene volcanic rocks are distributed widely in this ore district. The biotite granite porphyry vein (12 Ma, Shi et al., 2017) is located in the central part of the Sinongduo deposit. Within the southeast part of the Sinongduo deposit, there is a granite porphyry pluton with a U-Pb age of 62.9-59 Ma (Zhang et al., 2024). This Miocene biotite granite porphyry intrudes into the Paleocene granite porphyry and affects the main Ag-Pb-Zn ores (Fig. 2a).

The Ag-Pb-Zn ore bodies predominantly occur as veins and breccia columns (Fig. 2b). The ore types chiefly consist of hydrothermal breccia Pb-Zn ores, hydrothermal vein Pb-Zn ores, and high-grade Ag ores (Fig. 3). The sulfides are predominantly composed of sphalerite, galena, pyrite, and chalcopyrite (Fig. 4a–f), along with minor silver minerals (Fig. 4g–i). The silver minerals are mainly argentite, pearceite, enargite, pyrargyrite, acanthite, and rare native silver (Li et al., 2017). Notable among these, argentite is the most significant silver mineral in the Sinongduo deposit, primarily filling the margins of early-formed pyrite (Fig. 4g). Pearceite and pyrargyrite commonly react with the galena, forming a residual texture (Fig. 4g, h). Acanthite is mainly hosted in the Fe-/Mn-bearing carbonates or red jasper (Fig. 4i). The gangue minerals primarily comprise quartz, sericite, illite, chalcedony, and calcite, with minor epidote, muscovite, and jasper. Through the crosscutting relationships (Fig. 3) and optical microscopy (Fig. 4), three paragenetic stages have been distinguished: (I) the pre-ore stage, primarily characterized by pyrite and quartz; (II) the main metallogenic stage, which includes three substages (Fe-Cu-Zn, Pb-Zn, and Ag), producing large amounts of sphalerite, galena, pyrite, chalcopyrite, and minor Ag minerals; and (III) the post-ore stage, marked by calcite and quartz (Fig. 5). Widespread intense hydrothermal alteration occurs near the prominent mineralized veins, characterized by quartz, sericite, illite, and calcite assemblages (Yang et al., 2020). Hydrothermal alterations in the Sinongduo deposit are chiefly characterized by silicification and illitization, as reported by Guo et al. (2020). Montmorillonitization has been identified using the shortwave infrared techniques (Guo et al., 2019).

4. Sampling and analysis

4.1. Sampling

For this study, samples were gathered from diverse drill holes within the Sinongduo deposit, as depicted in Fig. 2b. Subsequently, these samples were fabricated into polished thin sections for further analysis. More than 80 sections were prepared and investigated using both reflected and transmitted light microscopy techniques. Micrographs for every sample were obtained at various magnifications to identify appropriate regions, as shown in Fig. 4. Sections representing the main metallogenic stage were chosen for subsequent in *situ* major and trace element analyses.

4.2. In situ LA-ICP-MS analysis

In *situ* trace element analysis of sulfides via Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) was conducted at the Guangzhou Tuoyan Analytical Technology Co., Ltd. in Guangzhou, China. The analysis utilized a NWR 193 nm ArF Excimer laser-ablation system interfaced with an iCAP RQ ICPMS. The ICPMS was calibrated with the NIST 610 standard glass to minimize oxide production. The system used 0.7 l/min He carrier gas, which was later mixed with 0.89 l/min Ar make-up gas. The laser was set at a fluence of 3.5 J/cm^2, operating at 6 Hz with a 30 μm spot size. Each analysis lasted 45 seconds, followed by a 40-second background measurement. The measured trace elements include ^{47}Ti, ^{51}V, ^{53}Cr, ^{55}Mn, ^{57}Fe, ^{59}Co, ^{60}Ni, ^{65}Cu, ^{66}Zn, ^{69}Ga, ^{72}Ge, ^{75}As, ^{82}Se, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{125}Te, ^{197}Au, ^{205}Tl, ^{208}Pb, and ^{209}Bi. Trace element compositions were calibrated by employing reference materials, specifically NIST SRM610 and MASS-1, following the method described by Liu et al. (2008). To evaluate the quality of the data, the sulfide pellet standards NIST SRM612 and GE8 were analyzed and we use the major element determined by EPMA as internal standard to calibrate the LA-ICP-MS data such as Zn for sphalerite, Fe for pyrite, and Pb for galena. The approaches and relevant considerations in such analyses have been detailed in previous studies by Wilson et al. (2002) and Danyushevsky et al. (2011). For more details on the analytical techniques, refer to Zhang et al. (2022). The trace element analytical results are presented in the Supplementary Data (Tables 1–3) and Figs. 6–8.

4.3. Principal component analysis

We employed principal component analysis (PCA) implemented through Origin 2017 software to conduct a comprehensive analysis of trace elements within sphalerite, pyrite, and galena. For the compositional data, the closure problem should be a major concern when dealing with geochemical data sets (Makvandi et al., 2016; Grunsky et al., 2020). Before the PCA analysis, we used the log ratio transformation method (Grunsky et al., 2010) in Origin 2017 software to process the LA-ICP-MS data of sulfide, and preprocessed the element data below the detection limit with the imputation method (Grunsky et al., 2017), so as to avoid the composite closure during the PCA analysis. The focus was on a specific set of trace elements: Fe, Mn, Co, Ni, Cd, Cu, Pb, In, Sn, Se, As, Ag, Sb, and Ga. The outcomes of the PCA, incorporating loading plots, score plots, and variance explanations of the principal components, are depicted in Figs. 9 and 10 (Wu et al., 2023).

5. Results

In this study, a total of 204 LA-ICP-MS spots were analyzed for trace elements, distributed as 60 in sphalerite, 86 in pyrite, and 58 in galena. Tables 1–3 list their major and trace element compositions. Figure 7 illustrates the concentration ranges of the targeted trace elements within sphalerite. Representative time-resolved element profiles for sphalerite, pyrite, and galena are depicted in Fig. 8. Common elemental behaviors across the three mineral phases are presented in Figs. 9 and 10. Collectively, these figures provide insights into elemental distributions, temporal variations, and mineral-specific trends across the analyzed samples.

5.1. Trace elements in sphalerite

Sphalerite matrix hosts preferential enrichment of Fe, Cd, Cu, Mn, Ag, Pb, and Sn relative to other analyzed elements (Table 1; Fig. 6a). Fe, the most abundant trace element, shows concentration ranges consistent with typical epithermal Ag-Pb-Zn deposits (Fig. 7). Time-resolved depth profiles reveal Fe distributions characterized by uniform, smooth signals across analyzed samples (Fig. 8a–c). Positive correlations are observed between Fe and Mn, Ag, Sb, Sn, and Ga (Fig. 11). Cd concentrations rank second highest among the analyzed elements, also displaying flat and consistent distributions in depth profiles (Fig. 8a–c). Cu exhibits significant variability, while Mn concentrations align with typical epithermal Pb-Zn deposits (Fig. 7), maintaining uniform signal profiles (Fig. 8a–c). Ag contents are consistent with epithermal deposit characteristics (Fig. 7). Sn concentrations reflect typical epithermal Pb-Zn deposit values and exceed those of other deposit types (Fig. 7). Pb shows broad variability, while In demonstrates concentrations comparable to epithermal deposits (Fig. 7) with parallel, flat signals relative to Zn (Fig. 8a–c). Time-resolved profiling further highlights distinct behaviors for Ag, Pb, Cu, and Sn, characterized by both stable baseline levels and synchronous fluctuations (Fig. 8b–c).

Sphalerite demonstrates minor concentrations of Co, Ga, As, and Sb. Co exhibits concentration ranges from 0.1 to 49.2 ppm, with uniform, smooth distributions aligned with Zn profiles across analyzed samples (Figs. 8a–c). Sb concentrations are consistent with typical epithermal Pb-Zn deposit signatures (Fig. 7). As spans 0.1–148.0 ppm, while Ga displays limited variability with flat, consistent signal profiles parallel to Zn (Figs. 8a–c). Collectively, these elements show restricted enrichment compared to major trace components in sphalerite, maintaining stable distributions across all analyzed samples.

Principal component analysis (PCA) was employed in this investigation to evaluate trace element datasets from sphalerite, encompassing Ag, Fe, Cu, Mn, Co, Sn, Pb, Sb, Ga, In, and Cd (Fig. 9). The PCA results, projected onto the PC1-PC2 plane, account for 45.0% cumulative variance in sphalerite samples. Elemental distribution patterns indicate that PC1 is the dominant component influencing most trace elements, with three distinct elemental associations identified: Mn, Ag-Cu-Fe, and Pb-Cd-In-Co-Sn-Sb-Ga (Fig. 9a, c). Specifically, Mn, Co, Sn, Pb, Sb, Ga, In, and Cd exhibit significant loadings on PC1 (26.1%), whereas Fe, Ag, and Cu demonstrate substantial contributions to PC2 (18.9%). These results highlight the primary controls on elemental distributions within sphalerite, with PC1 reflecting a combined influence of chalcophile and siderophile elements, and PC2 capturing redox-sensitive components.

5.2. Trace elements in pyrite

Pyrite from the Sinongduo deposit is characterized by elevated As and Pb concentrations (Table 2; Fig. 6b). Arsenic, the most abundant trace element in pyrite, exhibits concentrations spanning 2.0–12,060 ppm, with uniform, stable distributions aligned with Fe profiles (Figs. 8d–f). Lead concentrations display two distinct behaviors in time-resolved profiles: baseline stability (Fig. 8a) and synchronous fluctuations (Figs. 8b–c).

Minor elements in pyrite include Ga, Ge, In, and Ni. Silver concentrations range from 0.1–1,578 ppm, while Mn demonstrates variability between 0.1–203 ppm. Cobalt (0.04–562 ppm) maintains consistent, unvarying distributions parallel to Fe (Figs. 8d–f). Copper concentrations (0.5–139 ppm) show smooth profiles analogous to Fe (Figs. 8d–f), and Ni spans 0.6–33.1 ppm. Other trace elements, such as Ga, Cd, In, and Ge, occur at relatively low levels (Table 2). These results highlight pyrite's role as a repository for As and Pb, with minor contributions from other trace elements, maintaining stable distributions across analyzed samples.

PCA was employed to evaluate trace element datasets from pyrite, including Mn, Zn, Sn, Se, Pb, Sb, As, Ag, Cu, Co, and Ni (Fig. 10a). The PCA results, projected onto the PC1-PC2 plane, account for 47.3% cumulative variance in pyrite samples. Elemental distribution patterns indicate that PC1 serves as the dominant component influencing most trace elements, with four discrete elemental groupings identified: Zn-Mn-Sn, Pb-Se, Ag-As-Sb, and Cu-Co-Ni (Fig. 10a). Specifically, As, Ag, Cu, Co, Ni, Se, and Pb exhibit significant loadings on PC1 (27.6%), whereas Mn, Zn, Sn, and Sb demonstrate pronounced contributions to PC2 (19.7%). These results highlight the multivariate controls on elemental partitioning in pyrite, with PC1 reflecting redox-sensitive and chalcophile elements, and PC2 capturing siderophile and lithophile components.

5.3. Trace elements in galena

In galena, As, Ag, Se, Sb, and Fe show relatively high concentrations (Table 3; Fig. 6c). Notably, As has the highest content, ranging from 50.3 to 321.4 ppm, with flat signals akin to Pb (Fig. 8g–i). Ag follows, with contents from 30.1 to 223.7 ppm. Se contents in galena, higher than in sphalerite and pyrite, vary from 18.3 to 1,032 ppm. Sb ranges from 17.7 to 292.2 ppm, with signals parallel to Pb (Fig. 8g–i). Fe contents span four orders of magnitude, from 5.5 to 95,831 ppm. Cd varies from 1.1 to 16.9 ppm. Other elements like Co, Ni, Ge, In, Tl, and Bi are relatively low (Table 3).

PCA was applied to the trace element datasets of galena, incorporating Ag, As, Mn, Zn, Sb, Se, Cu, Fe, Sn, and Cd (Fig. 10b). The PCA results, projected onto the PC1-PC2 plane, explain 66.8% cumulative variance in galena samples. Elemental distribution patterns indicate that PC1 acts as the dominant component controlling most trace elements, with three discrete elemental groupings identified: Cu-Fe, Sn-Cd, and Ag-Zn-Mn-As-Se-Sb (Fig. 10b). Specifically, Cu, Fe, Ag, Zn, Mn, Se, As, and Sb exhibit significant loadings on PC1 (49.1%), whereas Sn and Cd demonstrate pronounced contributions to PC2 (17.7%). These results underscore the geochemical controls on elemental partitioning in galena, with PC1 reflecting a combined influence of chalcophile and redox-sensitive elements, and PC2 capturing Sn-Cd enrichment related to sulfide mineralization processes.

6. Discussion

6.1. Occurrence and substitution mechanisms of trace elements in various sulfides

The LA-ICP-MS has emerged as a transformative tool for deciphering geochemical signatures of trace elements in sulfide minerals, significantly advancing the understanding of occurrence and incorporation mechanisms (Sykora et al., 2018; Li et al., 2022; Xing et al., 2022; Hong et al., 2023; Zhao et al., 2023; Li et al., 2023; Wang et al., 2024). LA-ICP-MS depth profiling reveals distinct signal patterns: lattice-bound trace elements and nano-inclusions typically exhibit stable baseline profiles, whereas micro-inclusions manifest as transient signal peaks (Cook et al., 2009; Ye et al., 2011). Crystal-chemical compatibility, particularly ionic radius similarity, governs isomorphic substitution of divalent cations (e.g., Fe^{2+}, Mn^{2+}, Cd^{2+}) into sulfide structures (Keith et al., 2022; Yang et al., 2025). In contrast, trivalent/tetravalent elements (Sb^{3+}, In^{3+}, Tl^{3+}, Sn^{4+}, Ge^{4+}) often utilize charge-balanced substitution mechanisms, pairing with monovalent ions (Ag^{+}, Cu^{+}) or creating lattice vacancies to maintain electroneutrality (Frenzel et al., 2016; Gregory et al., 2015). Atomic ratio (AR) analysis has proven instrumental in distinguishing substitutional mechanisms in sulfides, with cation stoichiometry providing critical insights into defect chemistry and element partitioning (Johan, 1988).

6.1.1 Sphalerite

The time-resolved profiles of Fe, Mn, Cd, In, Ag, and other elements in sphalerite are smooth and flat, generally running parallel to the Zn signal. This suggests these elements likely exist as solid solutions or nano-inclusions within the sphalerite (Feng et al., 2022; Liu et al., 2023). Significantly, the spatial proximities of Fe, Ag, Cu, Cd, In, Ga, Sb, Sn, Pb, and Co in the loading plot (Fig. 9a) suggest comparable geochemical behaviors, potentially reflecting their incorporation into sphalerite via coupled substitution mechanisms. Conversely, the discontinuous peak-like profiles of Pb (Figs. 8b and c) indicate that micro-inclusions represent the dominant mode of occurrence for this element, rather than lattice-bound substitution.

Prior studies have established general agreement that Mn^{2+} commonly substitutes sphalerite via coupled substitution mechanisms, yet the nature of interactions between Mn^{2+} and Fe^{2+} remains a subject of debate. As illustrated in Figs. 9a and 11a, the similar distribution patterns in loading plots and significant positive correlation between Mn^{2+} and Fe^{2+} suggest analogous geochemical behaviors and potential synergistic incorporation into the sphalerite lattice. This relationship supports a plausible substitution model where Mn^{2+} + Fe^{2+} replace two Zn^{2+} cations ($Mn^{2+} + Fe^{2+} \leftrightarrow 2Zn^{2+}$), maintaining charge balance through combined divalent cation substitution.

The Fe vs. Ag (Fig. 11b), Cu vs. Ag (Fig. 11c), and Fe vs. Cu (Fig. 11d) diagrams exhibit linear relationships with slopes (K) approximating atomic ratio values: 1/2 (K = 2), 1/1 (K = 1), and 1/2 (K = 2), respectively. This correspondence indicates stoichiometric constraints where Fe/Ag = 1/2, Cu/Ag = 1/1, and Fe/Cu = 1/2. Collectively, these results suggest a charge-balanced substitution mechanism in sphalerite: $Fe^{2+} + 2Ag^{+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$.

Moreover, the positive correlations in the Sb vs. (Fe+Ag), Fe vs. (Ag+Ga), and Sn vs. (Ag+Fe) plots imply these elements also enter sphalerite via coupled substitution. The Fe vs. Sb (Fig. 11f) and Sb vs. Ag (Fig. 11g) plots also show linear trends, with slopes (K) close to ARs at 1/4 (K = 4) and 2/1 (K = 0.5), yielding atomic ratios of Fe/Sb and Sb/Ag as 1/4 and 2/1. Combined with the earlier Fe vs. Ag analysis (Fig. 11b), we propose another possible substitution mechanism in sphalerite: $Fe^{2+} + 2Ag^{+} + 4Sb^{3+} \leftrightarrow 8Zn^{2+}$.

Furthermore, the Fe vs. Ga (Fig. 11i) and Ga vs. Ag (Fig. 11j) plots exhibit linear distributions, with slopes (K) near the atomic ratios (ARs) of 1/2 (K = 2) and 1/1 (K = 1). The Fe/Ga and Ga/Ag atomic ratios are 1/2 and 1/1. Thus, the coupled substitution between Fe, Ag, Ga, and Zn could be $Fe^{2+} + 2Ag^{+} + 2Ga^{3+} \leftrightarrow 5Zn^{2+}$. The regression slopes for Sn vs. Ag and Fe vs. Sn are close to 0.5 and 4, making the ARs of Sn/Ag and Fe/Sn 2/1 and 1/4. Therefore, these observations suggest a possible coupled substitution mechanism involving Ag, Fe, Sn, and Zn: $2Ag^{+} + Fe^{2+} + 4Sn^{4+} \leftrightarrow 10Zn^{2+}$.

6.1.2 Pyrite

Time-resolved LA-ICP-MS profiles of pyrite (Fig. 8) reveal stable signal characteristics for As, Co, Cu, Ni, Zn, and Pb, suggesting their incorporation into the pyrite lattice as solid solutions. Conversely, Mn, Ag, Pb, and Sn exhibit synchronous fluctuations superimposed on stable baselines, indicating a dual mode of occurrence involving Mn-Ag-Pb-Sn micro-inclusions and lattice-bound substitution.

Extensive research has documented multiple coupled substitution mechanisms in pyrite, such as $Ag^{+} + (Sb^{3+}, As^{3+}) \leftrightarrow 2Fe^{2+}$ (Wu et al., 2023; Li et al., 2023) and $Ag^{+} + As^{3+} \leftrightarrow 2Fe^{2+}$ (Chouinard et al., 2005). In the loading plot (Fig. 10a), Ag, Sb, As, and Co are closely located, implying they might share similarities and enter the pyrite crystal structure through coupled substitution. The Ag vs. As (Fig. 12a) and Ag vs. Sb (Fig. 12b) plots are linear, with slopes (K) close to atomic ratios (ARs) of 1/1 (K = 1), indicating Ag/As and Ag/Sb atomic ratios of 1/1. However, although As vs. Sb shows a positive correlation, the scattered data points (Fig. 12c) make determining the As vs.

Sb AR impossible. Therefore, we suggest a pyrite substitution mechanism: Ag^{+} + (Sb^{3+}, As^{3+}) ↔ 2Fe^{2+}. Furthermore, the Ag vs. Co plot is linear, with a Co vs. Ag atomic ratio of 1/1 (Fig. 12d). The Co vs. As and Co vs. Sb data points show no linear correlations (Fig. 12e and 12f), so the ARs of Co vs. As and Co vs. Sb can't be determined. Based on this, we propose another coupled substitution mechanism for Ag in pyrite: Ag^{+} + Co^{2+} + (Sb^{3+}, As^{3+}) ↔ 3Fe^{2+}.

6.1.3 Galena

The profiles of Ag, Sb, and As in time-resolved analyses show smooth, flat curves parallel to Pb's signal (Fig. 8g–i), indicating their presence as solid solutions in galena. Cu, Zn, Ga, Sn, and In display parallel fluctuations and smooth signals (Fig. 8h and 8i), suggesting they exist as Cu-Zn-Ga-Sn-In inclusions and solid solutions. The geochemical covariation between Ag and Sb or As (Figs. 12g–h) suggests charge-balanced substitution mechanisms, with spatial proximity in loading plots and stoichiometric relationships consistent with coupled incorporation into the pyrite lattice (Cook et al., 2009). The Ag vs. Sb (Fig. 12g) and Ag vs. As (Fig. 12h) plots are linear, with slopes close to atomic ratios (ARs) of 1/1 (K = 1) and 1/2 (K = 2), giving Ag/Sb and Ag/As ARs of 1/1 and 1/2. However, the Sb vs. As plot lacks a linear correlation, making the Sb vs. As AR indeterminable. Two possible coupled substitution mechanisms in galena are proposed: Ag^{+} + Sb^{3+} ↔ 2Pb^{2+} and 4Ag^{+} + 2As^{3+} ↔ 5Pb^{2+}, similar to the previously proposed mechanism Ag^{+} + (As^{3+}, Sb^{3+}) ↔ 2Pb^{2+} (George et al., 2015; Wu et al., 2023).

6.2. Ore formation temperature

Previous research has demonstrated that certain trace elements in sulfides, such as Fe, Mn, Cd, In, and Co, can serve as indicators for assessing precipitation temperature (Belissont et al., 2016; Frenzel et al., 2016; Bauer et al., 2019; Zhou et al., 2022; Wu et al., 2023; Li et al., 2024). High-temperature sphalerite (200–355 °C) typically exhibits dark color and elevated concentrations of Fe (35,800–114,200 ppm) and Mn (2,000–4,000 ppm), whereas low-temperature sphalerite (110–180 °C) tends to be lighter in hue with reduced Fe (2,300–20,000 ppm) and Mn (30–500 ppm) contents (Cook et al., 2009; Frenzel et al., 2016). In the Sinongduo deposit, the Fe content in sphalerite ranges from 7,273 to 50,019 ppm (average 25,561 ppm), and the Mn content ranges from 43 to 1,949 ppm (average 293 ppm) (Table 1; Fig. 6). Additionally, sphalerite samples from the Sinongduo deposit display a moderate dark color (Fig. 3). The Zn/Cd and In/Ge ratios of sphalerite in the Sinongduo deposit are 40–448 ppm (average 117 ppm) and 3–1,052 ppm (average 92), respectively, which are comparable to those in intermediate-temperature sphalerites (average Zn/Cd = 100–500, average In/Ge = 50–100) (Ye et al., 2011).

The Sinongduo deposit contains numerous iron-bearing minerals, such as pyrite, indicating that Fe is a significant component of the ore-forming fluid. Using the empirical sphalerite thermometer formula Fe/Zn (sphalerite) = 0.0013*t-0.2953

(Keith et al., 2014), the calculated temperatures range from 235.8 to 286.3 °C, with an average of 257.4 °C. Additionally, the GGIMF geothermometer, which incorporates Ga, Ge, In, Fe, and Mn contents, can be used to calculate the sphalerite formation temperature (Frenzel et al., 2016, 2017). The formula for the GGIMFis thermometer for sphalerite is summarized as follows: PC1* = ln [m (Ga)$^{0.22}$ × m (Ge)$^{0.22}$ × m (Fe)$^{-0.37}$ × m (Mn)$^{-0.2}$ × m (In)$^{-0.11}$], t = -(54.4 ± 7.3) × PC1* + (208 ± 10) (where t represents the formation temperature of sphalerite, and m represents the molar content of the trace element) (Frenzel et al., 2016, 2017). The calculated formation temperature of sphalerite ranges from 226.0 to 288.0 °C, with an average of 256.8 °C (Table 1).

In summary, the Sinongduo Ag-Pb-Zn deposit sphalerite formation temperatures are constrained to a predominantly intermediate temperature range based on textural and compositional characteristics. Notably, sulfide mineralization temperatures derived from trace element thermometry in this study (197.2–280.8 °C; mean = 237 °C) are higher than previously reported fluid inclusion thermometry data in sphalerite (Li et al., 2019). This observed discrepancy may be attributed to methodological disparities in fluid inclusion analysis between gangue and ore minerals, as highlighted by Lüders (2017), which can lead to variable temperature estimates due to differences in trapping conditions and post-entrapment modification.

6.3. Genetic type

The trace element concentrations in sulfides such as sphalerite, pyrite, and galena vary across different genetic Pb-Zn deposit types, providing crucial information for distinguishing the genesis of mineral deposits (Gregory et al., 2015; Chu et al., 2022; Zhao et al., 2022; Li et al., 2024; Xiang et al., 2024). Sphalerites from skarn Pb-Zn deposits exhibit distinct enrichment patterns of Co, Mn, and In, accompanied by low abundances of Ga, Sn, and Cd (Wei et al., 2021; Li et al., 2023). Sphalerites from SEDEX Pb-Zn deposits exhibit significant enrichment in Mn and Sn, coupled with low concentrations of Cd, Ga, and Ge (Ye et al., 2011). Sphalerites from mississippi valley-type (MVT) Pb-Zn deposits exhibit enrichment in Cd, Ge, As, and Tl, while being depleted in Mn, Fe, In, Co, and Sn (Wu et al., 2023; Li et al., 2023). Sphalerites from volcanogenic massive sulfide (VMS) Pb-Zn deposits exhibit pronounced enrichment in In, Mn, and Sn, accompanied by low abundances of Ge and Ga (Ye et al., 2011). Epithermal Pb-Zn deposit sphalerites have Fe, Mn, Sn enrichment and Ge, Ga, In, Co depletion (Qi et al., 2022; Han et al., 2022). Sphalerite from the Sinongduo Ag-Pb-Zn deposit displays distinct enrichment patterns of Fe, Mn, Cu, Ag, Sn, and Pb, concurrently depleted in Ga, Ge, In, and Co (Fig. 3), a geochemical signature consistent with the epithermal type mineralization system. In the geochemical discrimination diagrams for sulfide minerals (Table 4 for the reference data in Fig. 13), including Fe vs. Mn (sphalerite, Fig. 13a), Mn vs. Co (sphalerite, Fig. 13b), Ag vs. Mn (sphalerite, Fig. 13c), (Co + Mn) vs. (Cu + Ag) (sphalerite, Fig. 13d), Co vs. Ni (pyrite, Fig. 13e), and Cu/Ag vs. Bi/Sb (galena, Fig. 13f), the Sinongduo sulfides mainly plot into the epithermal deposit field. This compositional consistency across multiple trace element ratios supports an epithermal genetic model for the deposit, aligning with previously documented elemental partitioning trends in shallow-level hydrothermal systems (Cook et al., 2009; Frenzel et al., 2016). In addition, Li et al. (2023) introduced a novel machine learning-based discriminant model using PCA, which is capable of distinguishing different types of Pb-Zn deposits by analyzing the key trace elements present in sphalerite. In the PCA-based discriminant diagram, all the score points associated with the Sinongduo sphalerite are plotted in the epithermal-type region (Fig. 9b). This spatial distribution of score values, derived from multivariate statistical analysis of Mn, Fe, In, Ga, and Cd concentrations, further supports the epithermal genetic interpretation established through textural and compositional evidence.

Cumulative genetic evidence strongly supports the Sinongduo Ag-Pb-Zn deposit as a typical epithermal system, encompassing multiple lines of geologic and geochemical data: (1) hypogene mineralization is predominantly hosted at shallow crustal levels (<1 km) in vein, stockwork, and disseminated styles (Tang et al., 2016); (2) mineral assemblages are dominated by sphalerite, galena, pyrite, argentite, pearceite, and minor chalcopyrite, consistent with epithermal sulfide paragenesis (Tang et al., 2016; Li et al., 2017; Yang et al., 2020); (3) ore-forming fluids exhibit predominantly low- to intermediate-temperature characteristics (185.6–290.4 °C) and dilute salinity (1.4–8.94 wt% NaCl equiv.), as constrained by fluid inclusion microthermometry (Li et al., 2019); (4) hydrothermal alteration assemblages are typified by sericitization, illitization, chalcedony, jasper, and quartz, reflecting shallow-level acid-leaching processes (Guo et al., 2018, 2019, 2020); (5) isotopic studies indicate a magmatic-hydrothermal fluid source, consistent with epithermal system evolution (Fu et al., 2017; Ding et al., 2017); and (6) high-precision geochronology documents temporal coincidence between mineralization (60.9–63.1 Ma) and coeval magmatic events (61.9–65.0 Ma), supporting a magmatic-related genetic model (Ding et al., 2017; Yang et al., 2020).

7. Conclusions

(1) Trace element occurrences in Sinongduo deposit sulfides are dominated by lattice-bound solid solutions and nanoscale inclusions, reflecting crystal-chemical compatibility and element partitioning dynamics during mineralization. Silver exhibits multiple substitutional incorporation mechanisms across sulfide phases: in sphalerite, charge-balanced reactions include $2Ag^+ + Fe^{2+} + 2Cu^+ \leftrightarrow 3Zn^{2+}$, $2Ag^+ + Fe^{2+} + 4Sb^{3+} \leftrightarrow 8Zn^{2+}$, $2Ag^+ + Fe^{2+} + 2Ga^{3+} \leftrightarrow 5Zn^{2+}$, and $2Ag^+ + Fe^{2+} + 4Sn^{4+} \leftrightarrow 10Zn^{2+}$; in pyrite, $Ag^+ + (Sb^{3+}, As^{3+}) \leftrightarrow 2Fe^{2+}$ and $Ag^+ + Co^{2+} + (Sb^{3+}, As^{3+}) \leftrightarrow 3Fe^{2+}$; and in galena, $Ag^+ + Sb^{3+} \leftrightarrow 2Pb^{2+}$ and $4Ag^+ + 2As^{3+} \leftrightarrow 5Pb^{2+}$.

(2) Geothermometric constraints derived from trace element thermometry yield temperature estimates of 226.0–288.0 °C (mean = 256.8 °C) for the principal mineralization stage at the Sinongduo deposit. These values are in close agreement with previously reported fluid inclusion homogenization temperatures (197.2–

280.8 °C), corroborating the intermediate-temperature nature of the hydrothermal system.

(3) The Sinongduo Ag-Pb-Zn deposit is interpreted as a typical epithermal system based on integrated geological and trace element signatures of sulfide minerals. Analyzing trace elements in sulfides using in *situ* LA-ICP-MS can help identify the genetic types of ore deposits.

Declaration of Competing Interest

The authors confirm that there are no known financial conflicts of interest or personal relationships that might influence the work reported in this paper.

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Figure captions

Fig. 1. (a) The Tibetan Plateau; (b) tectonic framework of the Tibetan Plateau (after Zhu et al., 2011); (c) the simplified diagram showing the distribution of the magmatic rocks and the associated deposits in the Lhasa terrane (modified from Lang et al., 2019). Abbreviations: YGL-Yaguila; SR-Sharang; DZL-Dongzhongla; HHG-Hahaigang; MYA-Mengya'a; TBL-Tangbula; JM-Jiama; DZSD-Dongzhongsongduo; BP-Bangpu; LML-Longmala; XGG-Xingaguo; LQL-Leqingla; QL-Qulong; CBP-Chengbapu; ZBL-Zhibula; LTG-Lietinggang; TG-Tinggong; CJ-Chongjiang; JR-Jiru; XC-Xiongcun; CGL-Chagele; QG-Qiagong; JDPL-Jiaduopule; LG-Longgen; ZN-Zhunuo; NRSD-Narusongduo; SND-Sinongduo.

Fig. 2. (a) A geological map of the Sinongduo Ag-Pb-Zn deposit and (b) the cross-section diagram showing the representative orebodies in the Sinongduo Ag-Pb-Zn deposit (modified from Zhang et al., 2024).

Fig. 3. Representative photographs and crosscutting relationships of the sulfide ores

from the Sinongduo Ag-Pb-Zn deposit. (a) The Explosive breccia type Pb-Zn ore in the rhyolite porphyry and crystal tuff; (b) the dense disseminated Pb-Zn ore; (c) the massive Pb-Zn ore; (d) the galena-sphalerite-quartz vein within the crystal tuff; (e) the early sphalerite-galena-quartz vein crosscut by the late quartz vein; (f) the early pyrite-quartz vein crosscut by the late quartz vein; (g) the early galena-sphalerite-pyrite-quartz vein within the rhyolite porphyry; (h) the early pyrite-quartz vein crosscut by the late quartz vein; (i) the early sphalerite-galena-quartz vein crosscut by the quartz vein and the late calcite vein crosscutting the quartz vein; (j) the early sphalerite-pyrite-quartz vein crosscut by the late quartz-calcite vein; (k) the late quartz vein crosscutting the early sphalerite-galena-pyrite vein; (l) the late quartz vein crosscutting the galena-sphalerite-quartz vein in the rhyolite porphyry. Abbreviations: Q-quartz, Cal-calcite, Sp-sphalerite, Gn-galena, Py-pyrite.

Fig. 4. Representative photomicrographs (reflected light) of the sulfide ores from the Sinongduo Ag-Pb-Zn deposit. (a) The early subhedral pyrite within the quartz; (b) the early subhedral pyrite replaced by the late galena and sphalerite; (c) the anhedral sphalerite with chalcopyrite solid resolution and coarse-grained subhedral chalcopyrite; (d-e) the sphalerite and galena; (f) the galena with typical triangular hole structure; (g) the mineral assemblage comprising argentite, pearceite, pyrite, galena, and pyrite; (h) the pyrargyrite surrounding the galena and the pyrite; (i) the akanthite mineral aggregates. Abbreviations: Arn-argentite, Pea-pearceite, Pyr-pyrargyrite, Aca-acanthite, Cpy-chalcopyrite, Sp-sphalerite, Gn-galena, Py-pyrite, Q-quartz, Jas-jasper.

Fig. 5. The mineral paragenetic sequences and mineralization stages of the Sinongduo deposit.

Fig. 6. The statistical box and whisker plots of the sphalerite (a), pyrite (b) and galena (c) from the Sinongduo Ag-Pb-Zn deposit.

Fig. 7. Box plots of 14 trace elements in sphalerite from the Sinongduo Ag-Pb-Zn deposit, showing the element variations for five types of Pb-Zn deposits (data for other type deposits from Cook et al., 2009; Penney et al., 2004; Lockington et al., 2014; Frenzel et al., 2016; Fan et al., 2020; Hu et al., 2021; Xing et al., 2022; Yang et al., 2022; Li et al., 2023; Wu et al., 2023).

Fig. 8. Time-resolved LA-ICP-MS signal depth profiles for the representative sulfide samples from the Sinongduo Ag-Pb-Zn deposit.

Fig. 9. The results of PCA for the concentrations of trace elements in sphalerite from the Sinongduo Ag-Pb-Zn deposit. (a) Elements plotted in PC1 versus PC2 with an explaining of 45.0%; (b) score plot of all points in PC1 and PC2 for trace contents of sphalerite from the Sinongduo deposit and the five different types of Pb-Zn deposits (epithermal, skarn, MVT, SEDEX, MVT) (Data are from Xing et al., 2021; Yang et al., 2022; Qi et al., 2022; Zhou et al., 2022; Niu et al., 2023; Li et al., 2023; Wu et al.,

2023); (c) scree plot of eigenvalues of the correlation matrix; (d) loadings of the principal components in sphalerite.

Fig. 10. The loading plot of principal component analysis of pyrite (a) and galena (b) from the Sinongduo deposit.

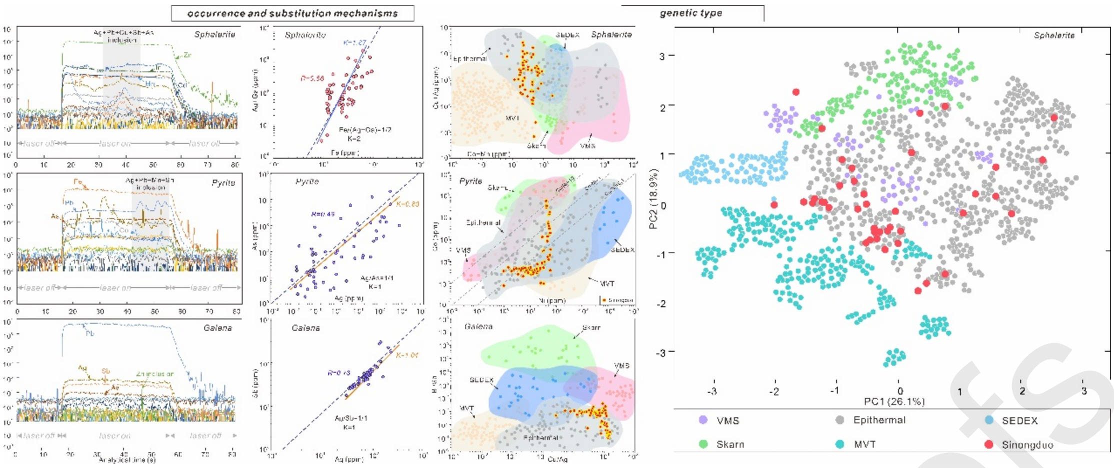
Fig. 11. The correlation plots of trace element in sphalerite. (a) Fe vs. Mn, (b) Fe vs. Ag, (c) Cu vs. Ag, (d) Fe vs. Cu, (e) Mx vs. My, (f) Fe vs. Sb, (g) Sb vs. Ag, (h) Fe vs. (Ag + Ga), (i) Fe vs. Ga, (j) Ga vs. Ag, (k) Sn vs. Ag, (l) Fe vs. Sn. Note: R = correlation coefficient; K = slope; lines in blue = regression lines; lines in black = theoretical WRs.

Fig. 12. The correlation plots of trace elements in pyrite (a-f) and galena (g-i). (a) Ag vs. As, (b) Ag vs. Sb, (c) As vs. Sb, (d) Ag vs. Co, (e) As vs. Co, (f) Co vs. Sb, (g) Ag vs. Sb, (h) Ag vs. As, and (i) Sb vs. As. Note: R = correlation coefficient, K = slope, lines in yellow = regression lines, lines in black = theoretical WRs, Py = pyrite, Gn = galena.

Fig. 13. Binary plots of trace elements in sulfides from the Sinongduo deposit and other genetic types of ore deposits (data are collected from Lockington et al., 2014; George et al., 2015; Hu et al., 2021; Liu et al., 2022; Zhou et al., 2022; Lin et al., 2023; Wu et al., 2023). (a) Fe vs. Mn, (b) Mn vs. Co, (c) Ag vs. Mn, and (d) (Co+Mn) vs. (Cu+Ag) for sphalerite, (e) Ni vs. Co for pyrite, and (f) (Cu/Ag) vs. (Bi/Sb) for galena.

Highlights

- A comprehensive LA-ICP-MS study was conducted on the Ag substitution mechanisms in sulfides from the Sinongduo deposit.
- The epithermal genetic type of the Sinongduo deposit was established through trace element geochemistry and thermometry analyses of the sulfides.
- In *situ* LA-ICP-MS trace element analyses of sulfides can help identify the genetic types of ore deposits.

1038 
occurrence and substitution mechanisms
genetic type
Sphalerite
Pyrite
Galena
laser off
laser on
PC1 (26.1%)
PC2 (18.9%)
VMS
Epithermal
SEDEX
Skarn
MVT
Sinongduo